Lab 1: Text Pre-processing

Information Retrieval (417524B) - SPPU BE AI&DS

Write a program for pre-processing of a text document such as stop word removal, stemming.

CODE

```
import nltk
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from nltk.tokenize import word_tokenize

text = ("The quick brown foxes are jumping over the lazy dogs while the "
        "children are playing happily in the beautiful green garden near "
        "their houses.")

tokens = word_tokenize(text)
print("Original Text:\n", text)
print("\nTokens:\n", tokens)

stop_words = set(stopwords.words("english"))
filtered = [w for w in tokens if w.lower() not in stop_words and w.isalnum()]
print("\nAfter Stop Word Removal:\n", filtered)

stemmer = PorterStemmer()
stemmed = [stemmer.stem(w) for w in filtered]
print("\nAfter Stemming:\n", stemmed)

# from nltk.stem import SnowballStemmer
# snow = SnowballStemmer("english")
# print([snow.stem(w) for w in filtered])
```

OUTPUT

```
Original Text:
 The quick brown foxes are jumping over the lazy dogs while the children are playing happily in the
beautiful green garden near their houses.

Tokens:
 ['The', 'quick', 'brown', 'foxes', 'are', 'jumping', 'over', 'the', 'lazy', 'dogs', 'while', 'the',
'children', 'are', 'playing', 'happily', 'in', 'the', 'beautiful', 'green', 'garden', 'near', 'their',
'houses', '.']

After Stop Word Removal:
 ['quick', 'brown', 'foxes', 'jumping', 'lazy', 'dogs', 'children', 'playing', 'happily', 'beautiful',
'green', 'garden', 'near', 'houses']

After Stemming:
 ['quick', 'brown', 'fox', 'jump', 'lazi', 'dog', 'children', 'play', 'happili', 'beauti', 'green',
'garden', 'near', 'hous']
```